Learning how to prove Workshop Week 11

Factor groups

$$
\begin{tikzcd}
G \arrow[r, "f"] \arrow[d, "\pi"'] & H \\
G/\ker(f) \arrow[ur, dashed, "\tilde{f}"'] &
\end{tikzcd}
$$

Normal subgroups. Factor groups

1. Let H be a normal subgroup in a group G such that $|H| = 2$. Show that $H \subseteq Z(G)$.

2. Let H be a subgroup of a group G such that $H \subseteq Z(G)$ and G/H is cyclic. Prove that G is abelian.